

3-Week Intensive Syllabus

Programme structure and assessment

SpaceCDF -- AI-Assisted Concurrent Design Facility

University of Ottawa - SEDTI

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DRAFT

Space Mission Design & Operations Programme

3-Week Intensive (Monday-Friday, 9:00-16:00)

Audience: Industry professionals, graduate students, upper-year undergraduates

Capacity: 20-30 participants in teams of 4-5

Tool: SpaceCDF AI-Assisted Concurrent Design Facility

WEEK 1: Canadian Space Landscape & Regulatory Environment

Day	Topic	Activities
Mon	The Canadian Space Ecosystem	Canadian Space Agency mandate & programs. Canadian space policy.
Tue	Regulatory Framework	ISED spectrum licensing (CPC-2-6-02). RSSSA (Remote Sensing of the Surface of the Earth Act).
Wed	International Standards & Compliance	ECSS standard framework and tailoring for small missions. ISO 9001.
Thu	Mission Needs & Opportunities	Identifying mission needs in the Canadian context (Arctic monitoring, Earth observation).
Fri	Space vs Non-Space Trade & Concept	Trade study methodology (criteria, weightings, scoring). Evaluation of mission concepts.

WEEK 2: Concurrent Design Facility -- Designing a Mission

Day	Topic	Activities
Mon	System-V & Requirements	System-V model deep dive. SMART requirements from objectives.
Tue	Subsystem Design: Power, AOCS, Thermal	Orbit selection trade. Solar array & battery sizing. Duty cycle.
Wed	Subsystem Design: Comms, Structure, Propulsion	Link budget calculation. Frequency band selection & licensing.
Thu	Integration, Verification, Cost	Interface matrix review. V&V matrix assignment. Environmental testing.
Fri	Design Review & Optimization	Run optimizer. Sensitivity analysis. Gate review (MCR criteria).

WEEK 3: Mission Operations & Simulation

Day	Topic	Activities
Mon	Operations Concept Development	Ground segment architecture. MCS setup (COSMOS/OpenM).
Tue	Mission Operations Training	Console positions and responsibilities. Communication protocols.
Wed	Mission Simulation Day 1	Full-day simulated LEOP + commissioning. Teams operate the mission.
Thu	Mission Simulation Day 2	Full-day simulated nominal operations + contingency. Science operations.
Fri	Wrap-Up & Presentations	Final design documentation review. Each team presents conclusions.

Assessment

Component	Weight	Description
Worksheets (daily)	20%	Completed during sessions, formative feedback
Design Review (Week 2 Fri)	30%	Team presentation of preliminary design
Simulation Performance (Week 3)	20%	Console operations, anomaly response
Final Presentation (Week 3 Fri)	30%	Complete mission design, ops concept, lessons

Prerequisites

- Engineering background (any discipline) or equivalent experience
- Basic familiarity with orbital mechanics concepts (recommended, not required)
- Laptop with modern web browser (Chrome/Firefox/Edge)

Materials Provided

- SpaceCDF tool access (web-based, hosted for the programme)
- Facilitator's Book (reference, post-programme access)
- Learner's Workbook (worksheets, tool guides)
- ECSS/NASA standard excerpts (relevant sections)
- Canadian regulatory filing templates

Outcomes

Participants will leave with:

- A complete CubeSat mission design (documented, reviewed, ECSS-compliant)
- Hands-on experience with concurrent design methodology
- Understanding of Canadian space regulatory environment
- Mission operations experience including anomaly response
- Network of peers across industry and academia